Problem Set 3: Instrumental Variables

MY457 | WT 2026 | Week 7

Add your answers to this .qmd file by replacing the dedicated placeholder text. When you are finished, render the .qmd file as a .pdf and make sure you push your work to GitHub.

1 Concepts

1.1 In the table below, fill in the missing values and **bold** the values you fill in. In one case, this is not possible. Which one is it and why? Note that Y_{1i} and Y_{0i} are the potential outcomes with respect to encouragement Z, so that $Y_i = Y_{Z_i,i}$.

Table 1: Table of Potential Outcomes

i	Z_i	D_{0i}	D_{1i}	D_i	Compliance Type	Y_{0i}	Y_{1i}	Y_i
1	0	**0**	**0**	**0**	Never-taker	3	3	**3**
2	1	0	1	**1**	**Complier**	2	5	**5**
3	1	1	1	**1**	**Always-taker**	4	4	**4**
4	0	1	**1**	**1**	**Defier**	3	0	3
5	0	1	**1**	1	**Always-taker**	**4**		4

1.2 Write down the estimand for the ITT. Excluding the unit for which you can't fill in the missing values, calculate the ITT using the potential outcomes in Table 1.

Delete this line and replace it with your answer.

1.3 In this example, is the ITT equal to the ATE (where the ATE is defined with respect to treatment D)? Why or why not?

Delete this line and replace it with your answer.

1.4 Beyond SUTVA, what must we assume to identify the LATE? From the data in Table 1, can you determine whether these assumptions hold? Why or why not?

Delete this line and replace it with your answer.

2 Simulation

2.1 Explain the code below and relate it to an instrumental variables data generating process. Calculate the true ITT and ATE.

```
set.seed(123)
n_obs <- 1000

data <- tibble(
    U = rbinom(n_obs, 1, .75),
    c_type = ifelse(U == 1,
        sample(1:3, n_obs, prob = c(0.7, 0.1, 0.2), replace = T),
        sample(1:3, n_obs, prob = c(0.35, 0.3, 0.35), replace = T)
    ),
    tau = case_match(c_type, 1 ~ 5000, 2 ~ 1000, 3 ~ 2500),
    Z = rbinom(n_obs, 1, .5),
    D = case_match(c_type, 1 ~ as.integer(Z == 1), 2 ~ 1L, 3 ~ 0L),
    Y0 = rnorm(n_obs, mean = 50000, sd = 2500) + 25000 * U,
    Y1 = Y0 + tau,
    Y = ifelse(D == 1, Y1, Y0)
)
```

Delete this line and replace it with your answer.

2.2 Create a DAG that represents the data generating process in the code above. Include the variables Z, C (for `c_type`), D, Y, and U.

Delete this line and replace it with your answer.

2.3 Use OLS to estimate the difference-in-means with respect to treatment D. Is this an unbiased estimator for the ATE in this setting? Why or why not?

Delete this line and replace it with your answer.

2.4 Now estimate the difference-in-means with respect to encouragement Z. Is this an unbiased estimator for any causal effect in this setting? Why or why not?

Delete this line and replace it with your answer.

2.5 Estimate the LATE using the Wald estimator.

Delete this line and replace it with your answer.

2.6 Using the Two Stage Least Squares (2SLS) estimator, re-estimate the LATE. Do this both manually (using two `lm()` commands) and using the `AER::ivreg` command. How does your result compare to the previous result? Make sure to present the results of your second stage in a well-formatted table that includes standard errors. Interpret the results and potential differences between the two models.

Delete this line and replace it with your answer.

3 Replication

In this section, we will use real-world data to reinforce what we have learned. We will analyse the dataset employed in *Foreign Aid, Human Rights and Democracy Promotion: Evidence from a Natural Experiment*.

Many countries, especially lower- and middle-income countries, are provided foreign aid with the intention that this will improve the living conditions of those in need. This foreign aid might have some spillover effects by encouraging the protection of human rights and entrenching democratic institutions. Does foreign aid improve human rights and democracy?

The authors use instrumental variables to study the effect of foreign aid, measured as overseas development assistance (`aid`) on the the CIRI Empowerment Index (`CIRI`).

3.1 Explain in your own words the instrumental variable (IV) design that the authors use to answer the research question. What are the instrument, the treatment, and the outcome?

Delete this line and replace it with your answer.

3.2 Give one example for a potential exclusion restriction violation in this setting. How plausible do you think the exclusion restriction is in this case?

Delete this line and replace it with your answer.

3.3 What would have to be true for the ignorability assumption to be violated in this setting? Do you think the ignorability assumption holds?

Delete this line and replace it with your answer.

3.4 Read into `R` the replication data set (`foreign_aid.dta`) and assign it to an object called `aid_data`. Each row in the data is a country-year observation. Estimate the ITT on the CIRI Empowerment Index (`CIRI`), controlling for country and year fixed effects. Be sure to present your results neatly, showing only relevant statistics for `Colony` and not for all your fixed effects. What do you find?

Delete this line and replace it with your answer.

3.5 Test whether the instrument is relevant. Use the `eff_F()` function from the `ivDiag` package. Your model should instrument Aid (`aid`) with presidency status (`Colony`), include country and year fixed effects as controls, and cluster standard errors by country and year. As a rule of thumb, an F-statistic greater than 10 is often considered to indicate a relevant instrument (Andrews, Stock, and Sun, 2019). What do you find?

Delete this line and replace it with your answer.

3.6 Estimate the first stage and the second stage, again controlling for two-way fixed effects. Again, use `lm()` and present the results neatly. What do you find?

Delete this line and replace it with your answer.

3.7 Using the `AER` package, use the `ivreg` function and estimate the LATE. Interpret the LATE substantively, are your results the same as above? Why?

Delete this line and replace it with your answer.

3.8 Estimate the LATE using the plug-in estimator. What do you find? Are your results different from before? Why?

Delete this line and replace it with your answer.

Feedback

Feedback to the problem set will be provided here.